Limnol. Oceanogr., 42(5, part 2), 1997, 1023–1038
© 1997, by the American Society of Limnology and Oceanography, Inc.

Aureococcus anophagefferens: Causes and ecological consequences of brown tides in U.S. mid-Atlantic coastal waters

V. Monica Bricelj[1] *and Darcy J. Lonsdale*
Marine Sciences Research Center, SUNY at Stony Brook, Stony Brook, New York 11794-5000

Abstract

Aureococcus anophagefferens is a picoplanktonic alga that since 1985 has bloomed in coastal embayments of the western mid-Atlantic, ranging from Narragansett Bay, Rhode Island, to Barnegat Bay, New Jersey, with greatest incidence of recurrence in Long Island bays, New York. Blooms of this small alga, referred to as "brown tide," can persist for several months during late spring and summer at densities in excess of 1.0×10^6 cells ml^{-1}. They are not associated with anomalous chlorophyll *a*, dissolved oxygen, or inorganic macronutrient (N, P) levels. Meterologically induced reduced flushing rates, elevated salinities, and delivery of micronutrients (e.g. iron) from the watershed have been implicated in bloom initiation. Brown tides have had severe detrimental effects on the benthos, especially eelgrass (*Zostera marina*) and suspension-feeding bivalves, including bay scallops (*Argopecten irradians*) and blue mussels (*Mytilus edulis*). Adult bivalves experience sublethal effects (e.g. inhibition of clearance rates) at *Aureococcus* concentrations as low as $\sim2\times10^5$ cells ml^{-1} and mortalities at $\sim10^6$ cells ml^{-1}, attributed to toxicity of this microalga. Impacts of brown tide on zooplankton are less clear, but reduced egg production rates of copepods and reduced population growth rates of ciliates are documented at higher brown tide concentrations ($\geq1.0\times10^6$ cells ml^{-1}). We summarize the state of knowledge about the physical, chemical, and biological factors that may contribute to brown tide initiation, maintenance, and decline and assess its ecological effects.

Summer algal blooms of a small (~2-μm diam) picoplankter, *Aureococcus anophagefferens,* referred to as "brown tide" due to the resulting water discoloration, have occurred since 1985 in several noncontiguous bays of the midwestern Atlantic coast of the U.S. These include Narragansett Bay, Rhode Island, the Peconics-Gardiners Bay system and south-shore bays of Long Island, New York, and New Jersey bays (Fig. 1). Brown tides have been associated with major deleterious ecosystem effects within this region, most notably on herbivorous grazers. The causative organism, *A. anophagefferens* Hargraves et Sieburth, a coccoid, nonmotile alga, has been classified as a chrysophyte (Sieburth et al. 1988). However, molecular phylogenetic studies support its inclusion within the Pelagophyceae, together with the species responsible for brown tides in the Laguna Madre-Baffin Bay ecosystem, Texas (DeYoe et al. 1995). Absorption and fluorescence spectra of *A. anophagefferens* are more typical of oceanic phytoplankton, such as the related chrysophyte *Pelagococcus subviridis,* thus suggesting an oceanic origin for this species (Yentsch et al. 1989). Despite widespread distribution of cells from the Gulf of Maine, to Great Bay, New Jersey, confirmed by immunofluorescent detection (Anderson et al. 1993), development of brown tides has been restricted to shallow, relatively unstratified estuaries (Fig. 1). They have not occurred in Long Island Sound, although *Aureococcus* cells are present at low levels in nearshore Connecticut waters (Anderson et al. 1993).

The first *Aureococcus* bloom occurred in summer 1985 and attained maximum densities of $0.9–1.5\times10^6$ cells ml^{-1} in Narragansett Bay (Sieburth et al. 1988; Smayda and Villareal 1989). In Long Island bays *Aureococcus* cell densities exceeded 2.5×10^6 cells ml^{-1} (Fig. 2, Nuzzi and Waters 1989), but we note that *A. anophagefferens* identification by immunofluorescence (Anderson et al. 1989) was not available in 1985; thus this early estimate may be inaccurate due to inadvertent inclusion of morphologically similar species. In Long Island bays brown tides have reappeared over the past decade with varying intensity, duration, and geographic spread since the first outbreak in 1985 (Fig. 2). Although spatial distribution can be patchy, *Aureococcus* cell densities during blooms are generally highest at the western end of the Peconic-Gardiners Bay estuary, and decline toward the eastern end (Nuzzi and Waters 1989). Bloom densities ($>0.5\times10^6$ cells ml^{-1}) have occurred at localized sites (e.g. confined bays such as West Neck Bay) in years when brown tide did not extend thoughout the estuary (Fig. 2). Widespread, high concentrations ($0.5–1.0\times10^6$ cells ml^{-1}) of *Aureococcus* cells recurred in summer 1995, with maxima $>1.7\times10^6$ cells ml^{-1} at some locations. Brown tide, presumed to have occurred in Barnegat Bay, New Jersey, in the mid-1980s (Olsen 1989), was verified in New Jersey coastal waters in 1995 (Fig. 1). Sieburth and Johnson (1989) identified *A. anophagefferens* as a minor component of Narragansett Bay plankton in water samples taken 3 yr before the 1985 brown tide. Although brown tide has not developed in Rhode Island waters since 1985, its presence has been repeatedly documented in experimental mesocosms fed Narragansett Bay water (Keller and Rice 1989; Nixon et al. 1994), suggesting that it may pose a threat to this ecosystem under favorable environmental conditions.

Aureococcus blooms typically develop in late May, attain

[1] Present address: Institute for Marine Biosciences, National Research Council, 1411 Oxford Street, Halifax, Nova Scotia B3H 3Z1.

Acknowledgments

We thank R. Nuzzi and R. Waters for providing unpublished data used in this review.

This paper is a result of research funded by NOAA award NA46RG0090 to the Research Foundation of SUNY for the New York Sea Grant Institute.

This is contribution 1057 from the Marine Sciences Research Center.

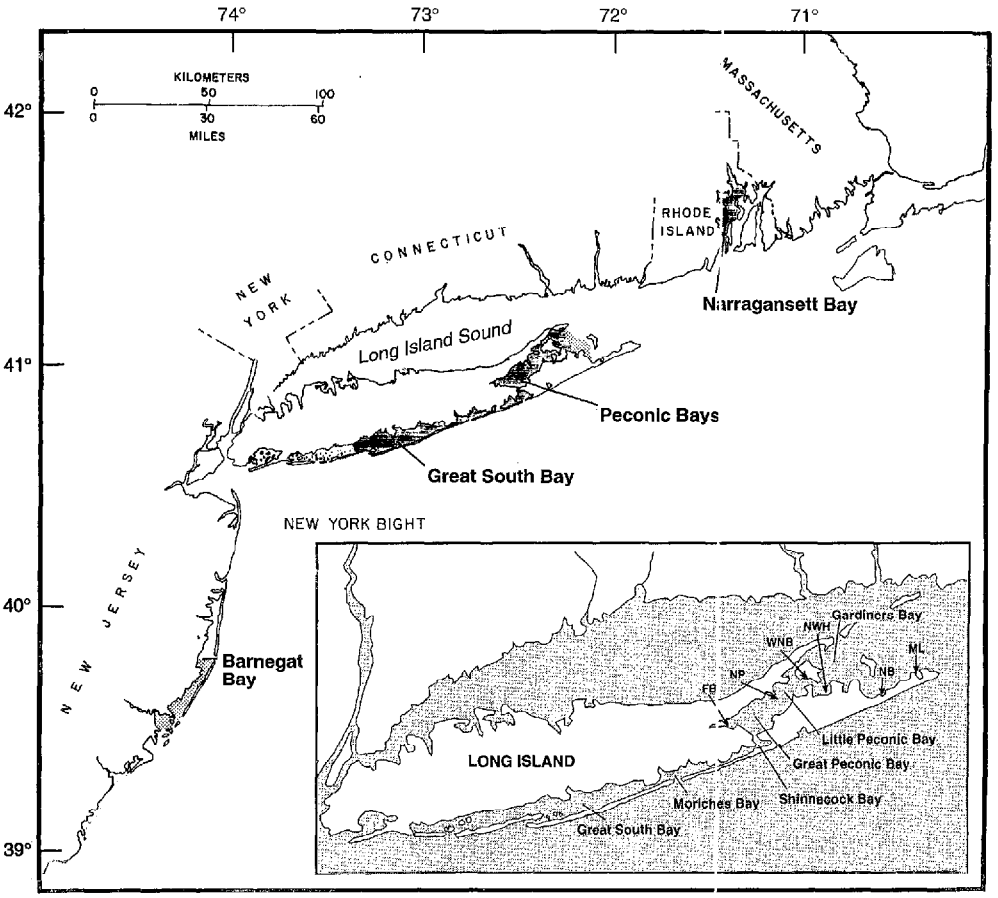

Fig. 1. Distribution of *Aureococcus anophagefferens* blooms [here defined by the occurrence of densities $>0.5\times10^6$ *Aureococcus* cells ml^{-1} (solid) or $>0.25\times10^6$ cells ml^{-1} (stippled)] along the mid-Atlantic coast of the U.S. Brown tides were reported in Barnegat Bay in 1985, 1986, and 1987, but not definitively attributed to *Aureococcus* (Olsen 1989) until 1995. Inset map shows Long Island study sites: FB—Flanders Bay; NP—Nassau Point; WNB—West Neck Bay; NWH—Northwest Harbor; NB—Napeague Bay; ML—Montauk Lake. Sources: SCDHS; Tracey 1988; Smayda and Villareal 1989.

peak densities in June or July, and wane in late summer (August–September) (Fig. 3). During brown tides, total primary productivity rates and phytoplankton biomass, as measured by chlorophyll *a* concentrations, remain at levels comparable to those of nonbloom years (Cosper et al. 1989*a*; Dennison et al. 1989). Blooms at their peak are virtually monospecific in some years and sites, e.g. in 1985 *Aureococcus* contributed >95% of total phytoplankton numbers in Narragansett Bay (Tracey et al. 1988; Sieburth et al. 1988), and >80% of total cellular phytoplankton volume throughout most of the bloom period in Peconic bays (Cosper et al. 1987). However, *Aureococcus* can also co-occur with significant numbers of other phytoplankton species at some locations (e.g. West Neck Bay in 1991, Lonsdale et al. 1996), during years of less intense brown tides (e.g. Peconic bays in 1987 and 1988) (Nuzzi and Waters 1989), or in later stages of the bloom (Smayda and Villareal 1989). The presence of *Aureococcus* in mixed vs. unialgal algal assemblages may have important implications in terms of mitigating its impact on grazers. The demise of the brown tide is associated with rapid replacement by other pico- and nanoplankton such as small diatoms and chlorophytes, which are present during brown tides, albeit in lower numbers compared to postbloom periods (Sieburth and Johnson 1989; Keller and Rice 1989). During the decline of the Long Island brown tide, rates of primary production in the <5-μm size-fraction did not change significantly relative to bloom conditions, also suggesting increased productivity of other nano- and picoplankton species (Cosper et al. 1989*a*). *Aureococcus* cells are present in the water column year-round, although sometimes in very low numbers (i.e. ~100 cells ml^{-1} in Long Island bays; Suffolk Co. Dept. Health Services, SCDHS). There is no evidence to date that brown tides are initiated from benthic cyst stages.

Mechanisms controlling bloom dynamics

Physical and chemical factors—The causal mechanisms of *A. anophagefferens* blooms remain largely hypothetical. The widespread but disjunct appearance of brown tide over

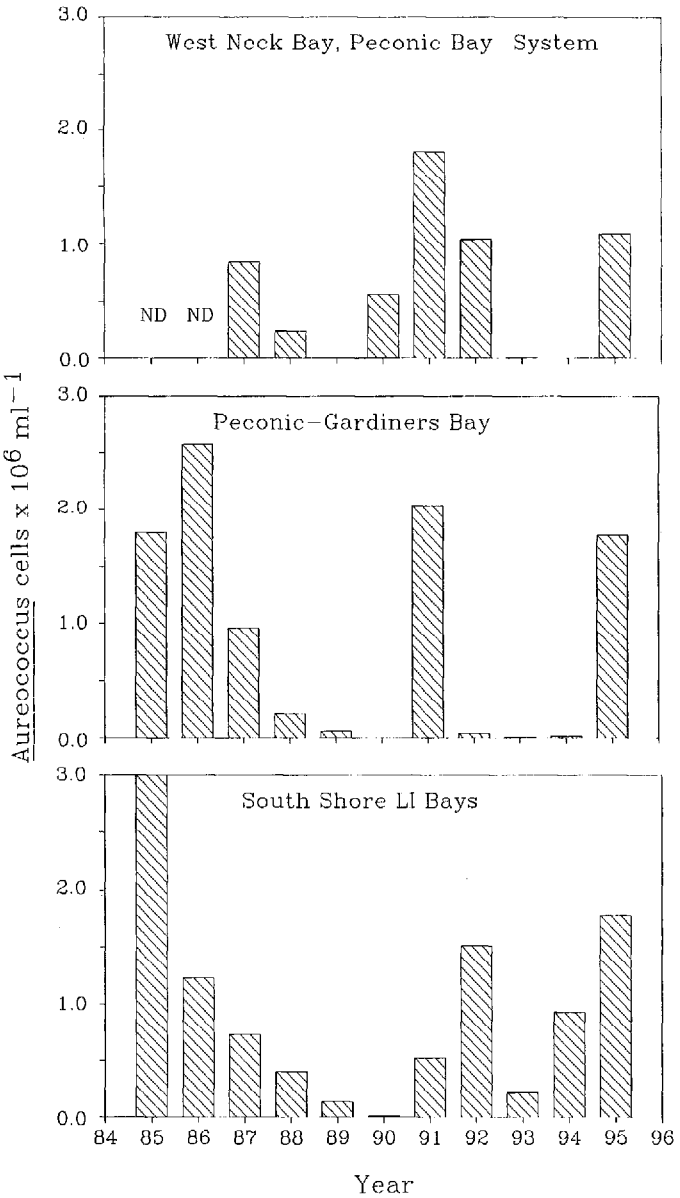

Fig. 2. Maximum annual concentration (cells ml^{-1}) of *Aureococcus anophagefferens* since the initial 1985 brown tide outbreak, in eastern and southern Long Island bays (*see Fig. 1*; ND—not determined). Counts since 1988 were determined by the immunofluorescent method and those in prior years by light microscopy (courtesy of SCDHS).

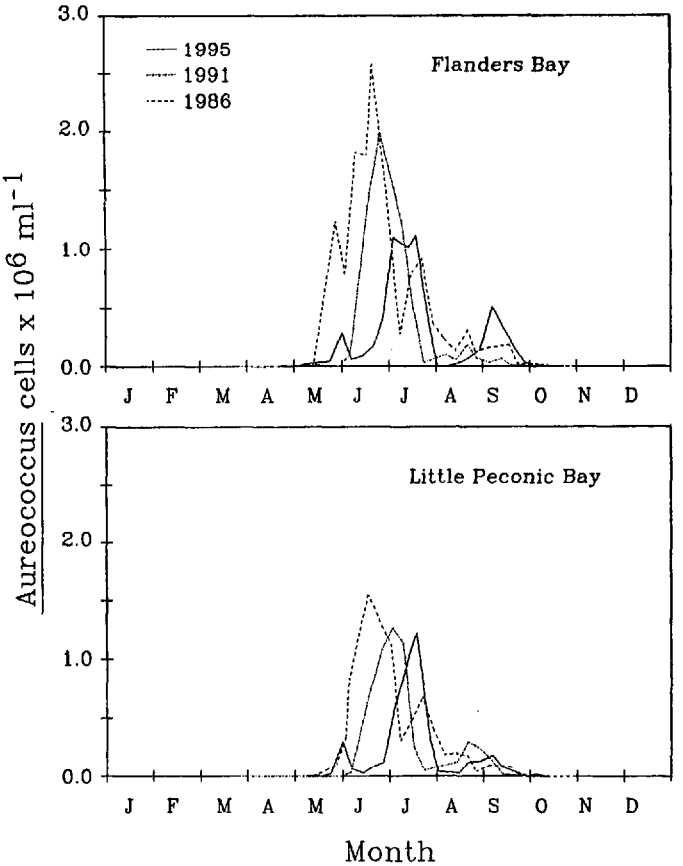

Fig. 3. Seasonal pattern in cell densities of *Aureococcus anophagefferens* during 3 yr of intense brown tides in Peconic bays, (*see Fig. 1*; weekly cell counts from SCDHS). (Note progressive delay in the timing of the peak throughout this period.)

a distance of ~500 km (Fig. 1) in 1985 led to the suggestion that it was triggered by mesoscale meteorological and (or) hydrographic events (Smayda and Villareal 1989; Cosper et al. 1989*b*). The onset of brown tide has been typically associated with reduced estuarine flushing rates and elevated salinities. In spring 1985, reduced flushing of southern Long Island bays was attributed to meteorological forcing (i.e. reduced wind stress and resulting low subtidal sea level oscillations; Vieira 1989; Vieira and Chant 1993). Blooms in the mid-1980s also followed anomalous winter and spring drought periods, characterized by rainfall levels well below the average of the previous four decades (Cosper et al. 1987). Drought conditions were also associated with the 1995 brown tide. Reduced rainfall markedly raised mean salinities in Long Island bays (from 25 to 30‰) and further contributed to the reduction in flushing rates. Unusually elevated salinities (31.5‰) were also reported during the 1985 Narragansett Bay brown tide (Keller and Rice 1989). Furthermore, the onset of blooms in the mid-1980s was associated with a pulse of heavy rainfall hypothesized to deliver specific micronutrients from the watershed, which favored cell growth of *Aureococcus* (Cosper et al. 1989*b*, 1990, 1993). Reduced flushing rates were thus suggested to promote retention of stimulatory micronutrients and allow buildup of *Aureococcus* populations within enclosed bays.

Although in the field *A. anophagefferens* occurs over a relatively broad salinity range (18–32‰, Anderson et al. 1993), field and laboratory studies strongly support the hypothesis that higher salinities (≥28‰) favor growth of *Aureococcus* populations. Most notably, during a widespread brown tide in Great South Bay in 1994, *Aureococcus* cell density was highly correlated with bay salinity, and blooms did not develop until salinities exceeded 27‰ (Gobler 1995). Maximum growth rates in culture (up to 0.8 d^{-1}) are achieved at 28–31‰ and decrease to negligible values be-

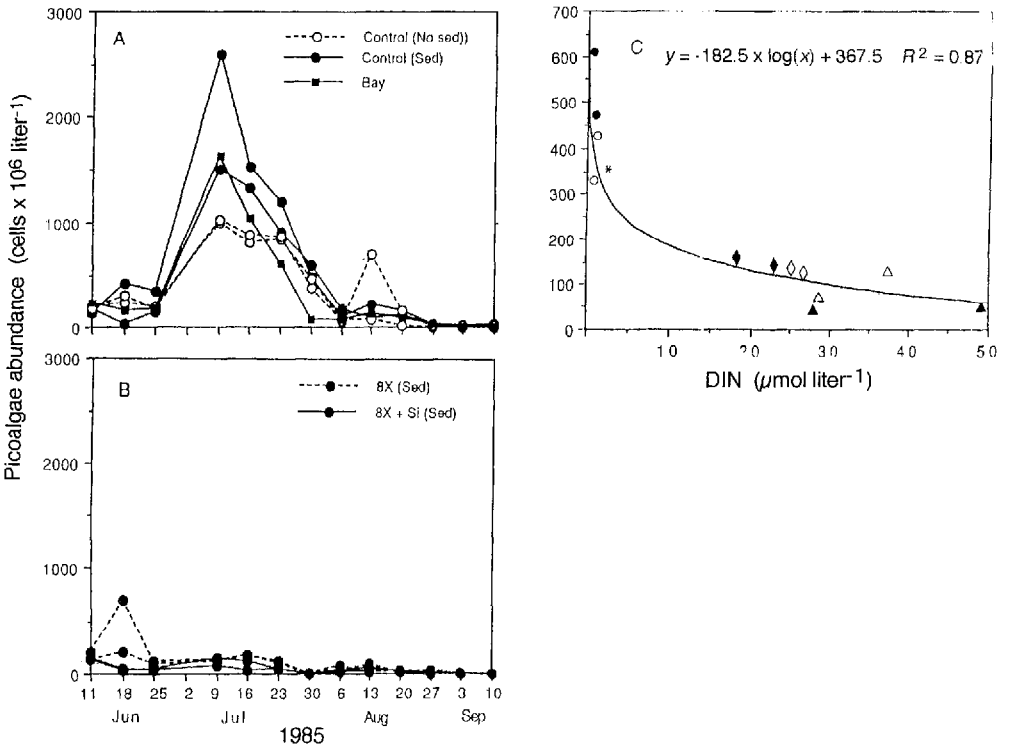

Fig. 4. A, B. Effect of macronutrient enrichment on the weekly cell density of Narragansett Bay picoalgae (predominantly *Aureococcus anophagefferens*) in experimental mesocosms, 1985 (from Keller and Rice 1989). A. Control mesocosms with and without bottom sediment, which included an intact benthic community (two replicate mesocosms per control). B. Nutrient-enriched mesocosms with sediment (comparable results were obtained without sediment). Nutrient levels: control (unenriched); 8×—nitrogen (as NH_4Cl), phosphate (KH_2PO_4), and silicate (Na_2SiO_3) at levels eight times the average annual loading to Narragansett Bay (two replicate mesocosms); 8× + Si—silicate loading at 112×. C. Negative relation between mean cell density of *A. anophagefferens* and mean dissolved inorganic nitrogen concentrations during the 1985 brown tide in Narragansett Bay and in experimental mesocosms (significant at $P < 0.05$). Treatment symbols: bay—*; controls-no sed.—○; controls-sed.—●; 8×-no sed.—◊; 8× + Si-no sed.—◆; 8×-sed.—△; 8× + Si-sed.—▲.

tween 21 and 25‰, depending on phosphate availability (Cosper et al. 1989*b*). Significant population growth of *A. anophagefferens* (0.4–0.5 d^{-1}) was maintained at salinities as low as 22‰ when glycerophosphate, instead of orthophospate, was added to culture media and cells were preadapted to the lower salinity (Cosper et al. 1989*b*).

Optimum temperature for growth of *Aureococcus* in culture ranges between 20 and 25°C (the maximum temperature tested), yielding growth rates of 0.8 and 0.6 d^{-1}, respectively (Cosper et al. 1989*b*). Thus, increasing water temperatures during late spring also seem to promote bloom development. Cultured *A. anophagefferens* cells, however, will grow at much lower temperatures (5°C), with a population doubling time of ~10d when preadapted to these temperatures (Cosper et al. 1989*b*). This ability to sustain slow growth at low temperatures allows seed populations to overwinter in mid-Atlantic waters.

A characteristic of *Aureococcus* that is likely to favor bloom development and persistence is its ability to acclimate to relatively low light levels and to maintain a high growth efficiency under "nutrient-saturating" conditions, at least up to 10^6 *Aureococcus* cells ml^{-1} (Milligan 1992) when compared with other algae. These features may render a competitive advantage to *A. anophagefferens* over other non-bloom-forming phytoplankton species.

Unlike the Texas brown tide alga, which is unable to use nitrate (NO_3^-) as a substrate for growth, *Aureococcus* can grow well in culture on either NO_3^- nitrite (NO_2^-), ammonium (NH_4^+) (DeYoe and Suttle 1994), or urea (Dzurica et al. 1989; Cosper et al 1990) as the sole nitrogen source. Inorganic macronutrient levels [NO_3^-, NO_2^-, NH_4^+ and phosphate (PO_4)] in Long Island bays during brown tide years did not differ significantly from prebloom years, and there was no correlation between PO_4 or dissolved inorganic nitrogen (DIN) concentrations and *Aureococcus* productivity (Cosper et al. 1989*a*). DIN concentrations immediately before onset of the 1985 Narragansett Bay brown tide also did not differ from pre- and postbloom years (Smayda and Villareal 1989). Furthermore, nutrient enrichment studies in mesocosms and field measurements in Narragansett Bay showed that mean *Aureococcus* cell density was inversely correlated with DIN concentration (Fig. 4C; Keller and Rice

1989; Smayda and Villareal 1989). In control mesocosms and concurrently in the bay, brown tide reached maximum values between 1 and 2.6×10^6 cells ml^{-1} in July and declined gradually through September (Fig. 4A). In contrast, in nutrient enrichment treatments (N, P, and Si with varying ratios of N : Si), blooms were relatively brief and remained at concentrations about an order of magnitude lower than in controls (Fig. 4B). Similarly, during summer 1992, *Aureococcus* attained densities of up to 5×10^5 cells ml^{-1} only in lagoonal mesocosms that contained low nutrient (N and P) levels, but not in nutrient-enriched treatments (Nixon et al. 1994). In summary, these studies strongly suggest that *Aureococcus* blooms do not occur in response to eutrophication (inorganic macronutrient loading). In fact, brown tide persistence has been related to its ability to grow at very low DIN levels—known to limit growth in other phytoplankters, including diatoms (Keller and Rice 1989). Nixon et al. (1994) further proposed that low flushing rates in Long Island bays in the mid-1980s (Vieira and Chant 1993) contributed to brown tide development by reducing the offshore input of nutrients (the major source of DIN to Great South Bay) rather than by increasing the retention of land-derived nutrients, as originally proposed by Cosper et al. (1989*b*).

Certain chelators and micronutrients, such as iron (Fe) and selenium (Se), have, however, been implicated as important factors influencing the onset of brown tides (Cosper et al. 1993; Gobler 1995; Gobler and Cosper 1996). Laboratory studies using defined culture media (Aquil) showed that both Fe and Se additions stimulate growth of *Aureococcus* populations (Cosper et al. 1993). Their effects are interdependent: only at higher Fe levels are the effects of Se concentrations apparent. The Fe requirement (Fe : C cellular ratio) for *A. anophagefferens* is higher than that of many other coastal phytoplankton species and comparable only to that of *Gymnodinium sanguineum,* a red tide dinoflagellate whose growth may be regulated by Fe availability (Gobler 1995; Gobler and Cosper 1996). Thus, *Aureococcus* cultures grown at <5 μM total dissolved Fe (100 nM labile Fe) experienced significantly reduced intracellular Fe content and growth rates. Growth enhancement of *Aureococcus* cells was achieved by adding Fe and Se to ambient seawater from bloom and nonbloom sites, suggesting that *Aureococcus* may at times experience in situ iron limitation (Cosper et al. 1993). Growth was stimulated in nonbloom water only if both elements were added (and only in low light conditions), whereas in bloom waters, Fe alone was sufficient for a positive growth rate response. Moreover, during brown tide development in West Neck Bay in 1992, dissolved Fe levels decreased at rates compatible with the calculated Fe demand of *Aureococcus* cultures and were replenished after the bloom subsided (Gobler and Cosper 1996). Dissolved Fe concentrations in Long Island bays are an order of magnitude higher than in other Atlantic estuaries and may therefore provide a suitable environment for brown tide (Gobler 1995). However, dissolved Fe levels are known to provide a poor measure of bioavailable Fe. Thus the extent to which Fe bioavailability may limit growth rates of *Aureococcus* during blooms and control the geographic distribution of brown tide in mid-Atlantic estuaries is unknown.

These findings, and a positive relationship between brown tide and dissolved Fe in Long Island waters in 1992, led to speculation (Schneider 1994) that the relatively recent occurrence of brown tides may be due to increased usage of deep groundwater, which ultimately enters bays as runoff or groundwater seepage and contains higher Fe concentrations than shallower aquifers. In this context, the effect of climate changes (e.g. drought periods) on groundwater flow and the mobilization and export of dissolved organic and inorganic micronutrients from deep groundwater merits consideration. Such climate-driven changes have been described for lake peatlands (Siegel et al. 1995).

Use of the chelators citric acid (CA) or nitrilotriacetic acid (NTA) in f/2 culture media (Guillard and Ryther 1962) was found to stimulate *Aureococcus* growth, relative to that of EDTA (Dzurica et al. 1989; Cosper et al. 1990). Such experimental results, in which the effects of chelators on algal growth are assessed in nondefined growth media, where complexation and bioavailability of essential and potentially toxic trace metals are unknown, are difficult to interpret unambiguously. Furthermore, CA could also stimulate growth by providing an organic carbon source for heterotrophic algae, thereby confounding experimental results. The ecological relevance of these results in the field remains undetermined, although they have led to speculation that increased usage of CA in lieu of phosphates in some detergents could play a role in the initiation of *Aureococcus* blooms (Cosper et al. 1993).

Biological factors—Competitive interactions with co-occurring phytoplankton, benthic and pelagic grazing, and control by viruses are recognized as potentially important biological mechanisms of control during algal blooms.

A. anophagefferens is both autotrophic and heterotrophic. Dzurica et al. (1989) showed that cultured cells displayed higher uptake rates per unit cell volume of ^{14}C- labeled organic compounds such as glutamic acid and glucose, both in the light and in the dark, compared to five other potentially co-occurring microalgae, including similar-sized species such as *Nannochloris* sp. and *Minutocellus polymorphus.* Although urea was not taken up by any of the algal species during the first 8 h of experimentation, *A. anophagefferens* did show some ability to takeup urea over longer time intervals. These results may be confounded by the presence of bacteria in nonaxenic *Aureococcus* cultures, which may also take up organic compounds. However, better growth of *Aureococcus* is achieved by adding specific organic nutrients to f/2 culture media, especially glycerophospate rather than orthophospate (Cosper et al. 1987).

Although the degree to which *Aureococcus* uses heterotrophic pathways in nature is unknown, a heterotrophic advantage could contribute to the formation and maintenance of brown tide, especially under light-limited conditions characteristic of *Aureococcus* blooms. *Aureococcus* may also be capable of extracellular, enzymatic oxidation of free amino acids and amines—a process that can contribute significantly to the total removal of amino acids (by oxidation plus heterotrophic uptake) in Long Island bays during summer (Pantoja 1992; Pantoja and Lee 1994) and result in ammonium incorporation into the cell (Palenik and Morel 1990). Relatively high rates of amino acid oxidation (up to 24 nM h^{-1}

by the nonbacterial size fraction) were measured during a brown tide in West Neck Bay (Pantoja 1992; Pantoja and Lee 1994). A capacity to produce inorganic N (NH_4^+) from organic compounds could provide *Aureococcus* with a competitive advantage relative to some phytoplankters such as *Synechococcus* (which shows no oxidative deamination activity) under conditions of inorganic N limitation.

Allelopathic effects of *Aureococcus* on other phytoplankton could explain its dominance during brown tides. Yet in a laboratory study of the effects of *A. anophagefferens* filtrate from cultured cells on common phytoplankton species (e.g. *Thalassiosira pseudonana, Prorocentrum minimum,* and *Nannochloris* sp.), little effect or enhanced growth of other algal populations at various filtrate concentrations (i.e. 0.1–100% concentration of the culture media) was shown (Cosper et al. 1989*b*). There is thus no evidence that *Aureococcus* filtrates have allelopathic properties. Interestingly, a 10% filtrate from senescent cells caused growth inhibition of *Aureococcus.* Several eucaryotic algae, e.g. the diatoms *M. polymorphus, T. pseudonana,* and *Skeletonema costatum,* co-occurred with *A. anophagefferens* in significant numbers (10^4–10^6 cells ml^{-1}) in August 1985 after the peak of the Narragansett Bay brown tide (Smayda and Villareal 1989). Dinoflagellates (*Dinophysis acuminata, Gymnodinium* spp., and the heterotroph *Polykrikos kofoidi*) were also relatively abundant during the 1991 brown tide in West Neck Bay (Lonsdale et al. 1996). Smayda and Villareal (1989) concluded that allelochemic regulation was probably not a significant deterrant of the 1985 brown tide. However, field and mesocosm studies (e.g. Keller and Rice 1989; Sieburth et al. 1988) found rapid replacement of *Aureococcus* cells by other picoplankton and nanoplankton after bloom cessation, and the usually dominant cyanobacteria *Synechococcus,* was depressed during the brown tide. Determining the mechanism(s) [i.e. changing physical-chemical conditions that may alter growth rate advantage among species, selective grazing and (or) reduction in allelopathy] that cause these phytoplankton community successional patterns warrants further research.

The development and persistence of brown tide may reflect failure of normal grazing control by zooplankton, especially protozoa and micrometazoa, or benthic suspension-feeders. It is unlikely that larger zooplankton, especially adult copepods, graze efficiently on *Aureococcus* due to its small size. Thus, during a brown tide in Narragansett Bay, *Acartia tonsa* adults had significantly lower gut pigments than those grazing on ambient phytoplankton enriched with the diatom *Thalassiosira weissflogii* (Durbin and Durbin 1989). In the summer months when brown tide appears, however, microzooplankton are the major consumers of phytoplankton in Long Island bays (Lonsdale et al. 1996; Mehran 1996).

In a laboratory study, Caron et al. (1989) found that two of five species of cultured protozoa grew in the presence of *Aureococcus* (at $\sim 10^6$ cells ml^{-1}), with or without an alternate bacterial food source. These protozoans, the microflagellate *Monas* sp. and a pleuronematid ciliate, consumed *Aureococcus* cells at rates equivalent to the natural growth rate of brown tide. Addition of cultured *Aureococcus* (at $\sim 2.5\times 10^5$ cells ml^{-1}) to a natural seawater sample taken from Vineyard Sound resulted in an increase in the density of protozoa, predominantly heterotrophic microflagellates, and concomitant decrease in the concentration of *Aureococcus.* Caron et al. (1989) also found no clear correlation between protozoan grazing rates on a fluorescently labeled chlorophyte and *Aureococcus* cell density (up to $\sim 5\times 10^5$ cells ml^{-1}) in Long Island bays, indicating that brown tide did not inhibit protozoan grazing on other microalgae. Field estimates of microbial consumption of *Aureococcus* cells were not obtained in this earlier study.

A 1995 grazing study in West Neck Bay using the dilution technique, also showed that microzooplankton exerted significant grazing pressure on total phytoplankton (measured as Chl *a* concentration) in the presence of similar brown tide concentrations (up to 2.5×10^5 cells ml^{-1}) when *Aureococcus* averaged 22% of total chlorophyll, but not when its contribution increased to 50% of the total algal biomass (Mehran 1996). That study also found that the microzooplankton community, primarily protozooplankton by numbers, was selective and generally avoided ingestion of *A. anophagefferens* cells. In laboratory experiments, the aloricate ciliate *Strombidium* sp., a dominant ciliate in Long Island bays, consumed *Aureococcus* cells (isolate CCMP1708 obtained from West Neck Bay in 1995) even when other microalgae (*Isochrysis galbana*) were present (Mehran 1996). It showed, however, a significantly higher electivity for *I. galbana* over brown tide cells, even when the latter was the dominant food item (measured by carbon concentration) during initial exposure to a mixed suspension. No selectivity was detected after several days of diet acclimation. Sieburth et al. (1988) also noted that phagotrophic protists, usually responsible for grazing on picoplankton, occurred in reduced numbers during the peak of the 1985 Narragansett Bay brown tide and were not observed feeding on *Aureococcus* cells until the bloom waned. In summary although some protozoans are capable of consuming *Aureococcus,* selective avoidance of *Aureococcus* by protozoans has been demonstrated both in the field and laboratory, suggesting that grazing inhibition of microzooplankton (protozoa and micrometazoa) is likely to contribute to brown tide initiation and maintenance.

Buskey and Stockwell (1993) showed that microzooplankton grazing impact on phytoplankton standing stocks was dramatically reduced, to <3% grazed per day compared to ~85–98% before the bloom of a related brown tide in Texas (DeYoe et al. 1995). Mesozooplankton such as adult *A. tonsa* showed little evidence of any phytoplankton grazing, as measured by gut-pigment analysis, during the bloom. Thus, for the Texas brown tide, grazing by zooplankton likely will not cause bloom cessation.

Although the impact of benthic grazers and their planktonic larvae on *A. anophagefferens* natural populations is also largely unknown, laboratory evidence indicates that bivalve grazing is strongly inhibited during brown tide. It is now recognized that, unlike most adults, bivalve larvae can effectively capture picoplankton-sized particles (Gallager 1988; Gallager et al. 1989, 1994). Yet bay scallop (*Argopecten irradians*) larvae showed low capture efficiencies for *Aureococcus* cells compared to other phytoplankton of similar size (e.g. *Nannochloris* sp.) in the laboratory (Gallager et al. 1989). Moreover, the presence of brown tide caused

inhibition of ingestion (but not capture) of other nutritious algal species due to increased particle rejection. Gallager et al. (1989) suggested that larval rejection behavior may be due to an exocellular compound produced by *Aureococcus* which renders all cells unpalatable or interferes with chemosensory perception. Inhibitory effects of brown tide on feeding of larvae of other bivalve species are unknown.

Grazing (clearance rates) of adult bivalves (mussels, *Mytilus edulis,* and northern quahogs, *Mercenaria mercenaria*) were markedly inhibited during the Narragansett Bay brown tide and provided early warning of an anomalous bloom event (Fig. 5; Tracey 1988). Particle retention efficiency of postmetamorphic bivalves generally declines exponentially at sizes <3–4 μm [5–7 μm for scallops (Pectinidae)] (Møhlenberg and Riisgard 1978). Thus, owing to their small size, *A. anophagefferens* cells were retained with only 59 and 36% efficiency by adult *M. edulis* and bay scallops, respectively, in short-term laboratory experiments (Cosper et al. 1987). High particle loading (algal biovolumes equivalent to 10^6 *Aureococcus* cells ml^{-1}) is also known to cause a marked increase in particle rejection and decrease in clearance rates and cell ingestion rates in bivalve larvae (Gallager 1988) and adults (Bricelj and Kuenstner 1989). However, clearance rates of adult bivalves on brown tide were much lower than on clay particles (Tracey 1988) or *Stichococcus* cells of comparable size at densities that stimulated bloom conditions (Tracey et al. 1988). Thus, reduced grazing on brown tide by adult bivalves cannot be solely ascribed to the small size and high density of *Aureococcus* cells. Whereas the dwarf surfclam, *Mulinia lateralis,* a sometimes dominant component of the Laguna Madre ecosystem, has the potential to exert grazing control on the Texas brown tide (Montagna et al. 1993), current evidence indicates that suspension-feeding bivalves such as mussels and scallops are unlikely to exert strong grazing pressure on *A. anophagefferens* once it has attained bloom densities ($\geq 10^5$ cells ml^{-1}).

Grazing avoidance of *Aureococcus* cells may result from production of toxic or inhibitory substances residing on the cell surface or within the cytoplasm. Bricelj and Kuenstner (1989) concluded that chronic toxicity and not small size, indigestibility, or nutritional deficiency contributed to juvenile and adult bivalve mortalities at bloom concentrations. In support of this hypothesis, in vitro studies demonstrated that the extracellular, diffuse, polysaccharidelike layer of *Aureococcus* cells contains a bioactive compound, released by amylase digestion, that is responsible for reduction in lateral ciliary beat frequency of isolated gills of some bivalves (Draper et al. 1990; Gainey and Shumway 1991). Its action could be mimicked by dopamine, but took longer (1–3 h of exposure) to take effect than the neurotransmitter. Because gill lateral cilia are involved in generating bivalve feeding currents, this work suggests a likely mechanism for the sharp reduction in clearance rates observed in adult bivalves during brown tide.

Using mussel clearance rates to assay for *Aureococcus* toxicity, Tracey et al. (1990) found preliminary evidence of variable toxicity of cells cultured under different temperature, light, and nutrient conditions. Although specific cell toxins associated with *Aureococcus* have not been identified, several studies (Tracey 1988; Ward and Targett 1989; Gallager et al. 1989; Gainey and Shumway 1991) agree in their finding that inhibitory effects on bivalve feeding require direct cell contact and are not elicited by dissolved metabolites present in cell-free filtrates of intact or lysed cells. *Aureococcus* also contains high levels of β-dimethylsulfoniopropionate (DMSP) per unit cell volume, comparable to those of other noxious algae such as *Phaeocystis* sp. and *Chry-*

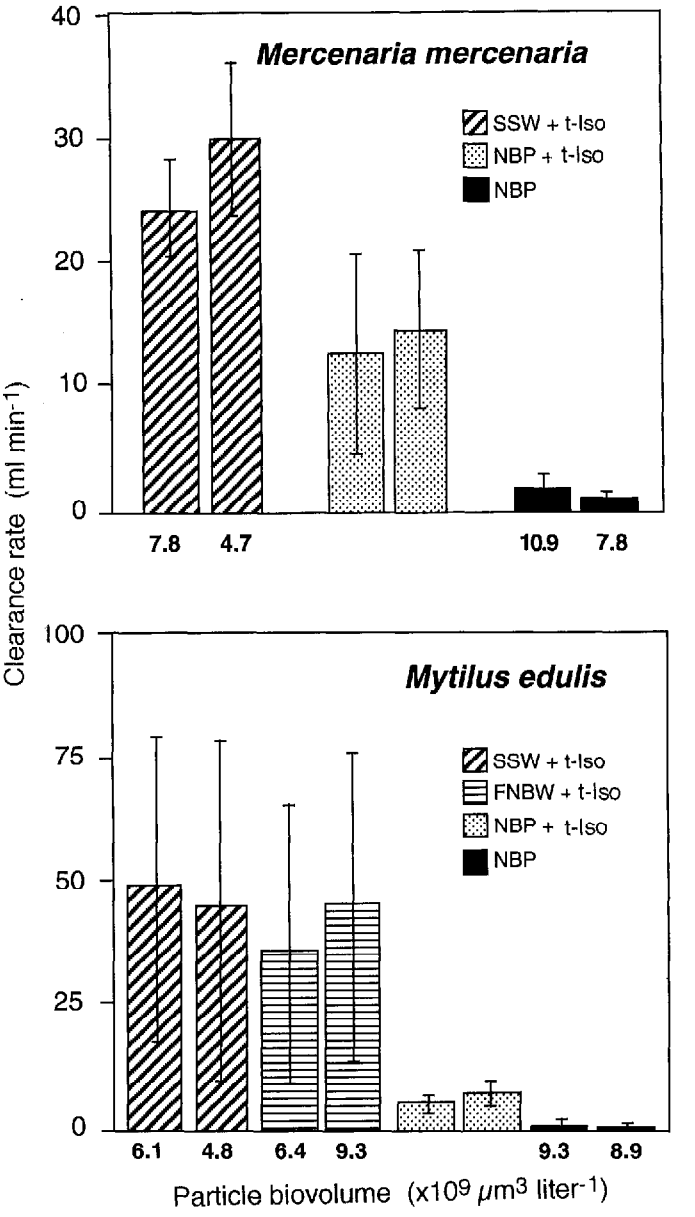

Fig. 5. Effect of *Aureococcus anophagefferens* on feeding rates (clearance rates) of adult quahogs (*M. mercenaria*) and blue mussels (*M. edulis*) (modified from Tracey 1988). NBP—ambient Narragansett Bay particles during the 1985 summer brown tide (*Aureococcus* attained 1.6×10^6 cells ml^{-1} and accounted for >95% of the total algal population: Sieburth et al. 1988); SSW—particle-free Sargasso seawater adjusted to ambient bay salinity; FNBW—0.45-μm filtered bay water to test for the effect of extracellular exudates of brown tide cells; t-Iso—cultured *Isochrysis galbana* (volume of NBP and t-Iso cells = 10.2 and 117.8 μm^3 cell^{-1}, respectively).

sochromulina polylepis (Keller et al. 1989). There is no evidence, however, that dimethylsulfide (DMS) or acrylic acid released during grazing and senescence of DMPS-containing algae are toxic to filter feeders.

Lysis of *Aureococcus* cells by viruses may also provide a natural biological control mechanism for brown tide. Widespread occurrence of virus particles in *Aureococcus* cells was observed in Narragansett Bay throughout the 1985 bloom (Sieburth et al. 1988) and in low-nutrient mesocosms in 1992 (Nixon et al. 1994), although their incidence did not correlate with bloom decline. Field observations of rapid dissipation of brown tide in some locations are consistent, however, with the rapid action of a viral agent, and the demise of laboratory cultures of *A. anophagefferens* has been observed following infection with a viral isolate from a Long Island bay (Milligan and Cosper 1994). The efficacy of this mechanism to control natural populations has yet to be demonstrated.

Ecosystem impacts

Benthic—The first notable impacts of brown tides in the mid-1980s were on the macrobenthos, namely eelgrass, *Zostera marina,* and bay scallops, *Argopecten irradians,* in Long Island bays (Cosper et al. 1987; Dennison et al. 1989; Bricelj et al. 1987) and mussels, *M. edulis,* in Narragansett Bay (Tracey 1988). Effects of brown tide on other benthic macrofauna, such as grazing gastropods, deposit-feeders, other filter-feeders, or predatory and scavenger species such as crabs and whelks, are unknown.

Brown tide coincides with the growth season of *Z. marina* in Long Island bays and results in severe light attenuation (50% reduction in mean Secchi disk depth during peak bloom conditions) (Dennison et al. 1989). Reduced light penetration is ascribed to the high density and enhanced light-scattering properties of small *Aureococcus* cells, rather than an increase in algal biomass. *Z. marina* has a relatively high minimal light requirement for survival (~20% of incident surface light, Dennison et al. 1993), and light availability is the primary factor limiting its depth distribution, biomass, and growth in shallow estuaries (Dennison 1987; Short et al. 1995). Outbreaks of brown tide in 1985–1986 caused significant reduction in depth penetration and leaf biomass of eelgrass in Peconic-Gardiners bays and Great South Bay (Cosper et al. 1987). This decline was not due to overgrowth and shading by epiphytic algae—responses commonly induced by nutrient enrichment (Dennison et al. 1989). However, anecdotal observations and limited historical data reviewed by Dennison et al. (1989) suggest that the effects of brown tide are superimposed on long-term reductions in eelgrass distribution in these two estuaries since the 1960s.

Eelgrass provides a critical nursery habitat and predator refuge for many finfish and benthic invertebrates, including bay scallops (Pohle et al. 1991). Therefore, loss of eelgrass may have contributed indirectly to poor recruitment of juvenile bay scallops and slow recovery of scallop stocks since the mid-1980s (Tettelbach and Wenczel 1993). Further work is needed to assess the long-term impact of brown tide-induced light attenuation on *Zostera* and its competitors, especially *Codium fragile,* a nonnative macroalga that has a lower light compensation point for growth than *Zostera* (Dennison et al. 1989) and is abundant in Peconic bays. A die-off of kelp populations (*Laminaria saccharina* and *Laminaria digitata*) occurred during the Narragansett Bay brown tide as a secondary effect of mass mussel mortalities, which led to loss of kelp attachment substrate in the euphotic zone (Smayda and Fofonoff 1989).

Although dissolved oxygen (DO) levels were not monitored during the 1985–1986 brown tide episodes in Long Island bays, they are unlikely to have reached levels detrimental to benthic organisms. Sedimentation of ungrazed *Aureococcus* cells and increased biological oxygen demand of bottom sediments would be precluded by this alga's small size, especially in shallow, weakly stratified bays such as Great South Bay and Peconic bays. No anomalous reductions in DO levels were recorded during a brown tide in West Neck Bay, a protected inner bay site, compared to non-brown tide years (Fig. 6). Furthermore, DO remained at relatively high levels (65–86% saturation) in Narragansett Bay bottom waters during the 1985 brown tide (Tracey 1988). Therefore, bivalve mortalities during brown tide cannot be attributed to near-bottom hypoxia-anoxia.

Aureococcus blooms, typically occurring in June–July, coincide with the period of spawning, planktonic larval development and juvenile growth of several commercially important bivalves in mid-Atlantic estuaries, thus threatening their reproductive success and early recruitment. Recruitment failure of bivalves during brown tides, documented for mussels and bay scallops, may be caused by gamete resorption in reproductive adults (suggested by Tracey 1988) or inability of an *Aureococcus* diet to support gametogenesis. It may also be caused by failure and(or) delay of larval settlement and metamorphosis and mortality or reduced growth, and thus greater vulnerability to predators, of postsettlement stages. The contributing role of these various processes to recruitment failure of natural populations is unknown.

Adverse effects of the brown tide on both adult and larval stages of suspension-feeding bivalves have been clearly established. In contrast to the Laguna Madre scenario, where the decline of *M. lateralis* preceded the brown tide and may have led to release of phytoplankton grazing pressure (Montagna et al. 1993), *A. anophagefferens* blooms preceded and are implicated as the direct cause of the subsequent demise of bivalve mollusc populations. Natural and transplanted populations of adult mussels (*M. edulis*) experienced mass (30–100%) mortalities in Narragansett Bay (Tracey 1988). Bay scallops (*A. irradians*) are semelparous (i.e. most adults do not survive to a second annual reproductive season). This short lifespan makes them particularly vulnerable to harmful algal outbreaks, as illustrated by the impact of the brown tide in Long Island bays, as well as that of the 1987–1988 bloom of *Gymnodinium breve* (=*Ptychodiscus brevis*) in North Carolina (Peterson and Summerson 1992). Laboratory studies showed that *A. anophagefferens* causes significant growth reduction and high mortalities of *A. irradians* larvae, even when present in a mixed suspension with a good algal food source (Fig. 7; Gallager et al. 1989). Adult bay scallops suffered severe reduction in adductor muscle weight (Bricelj

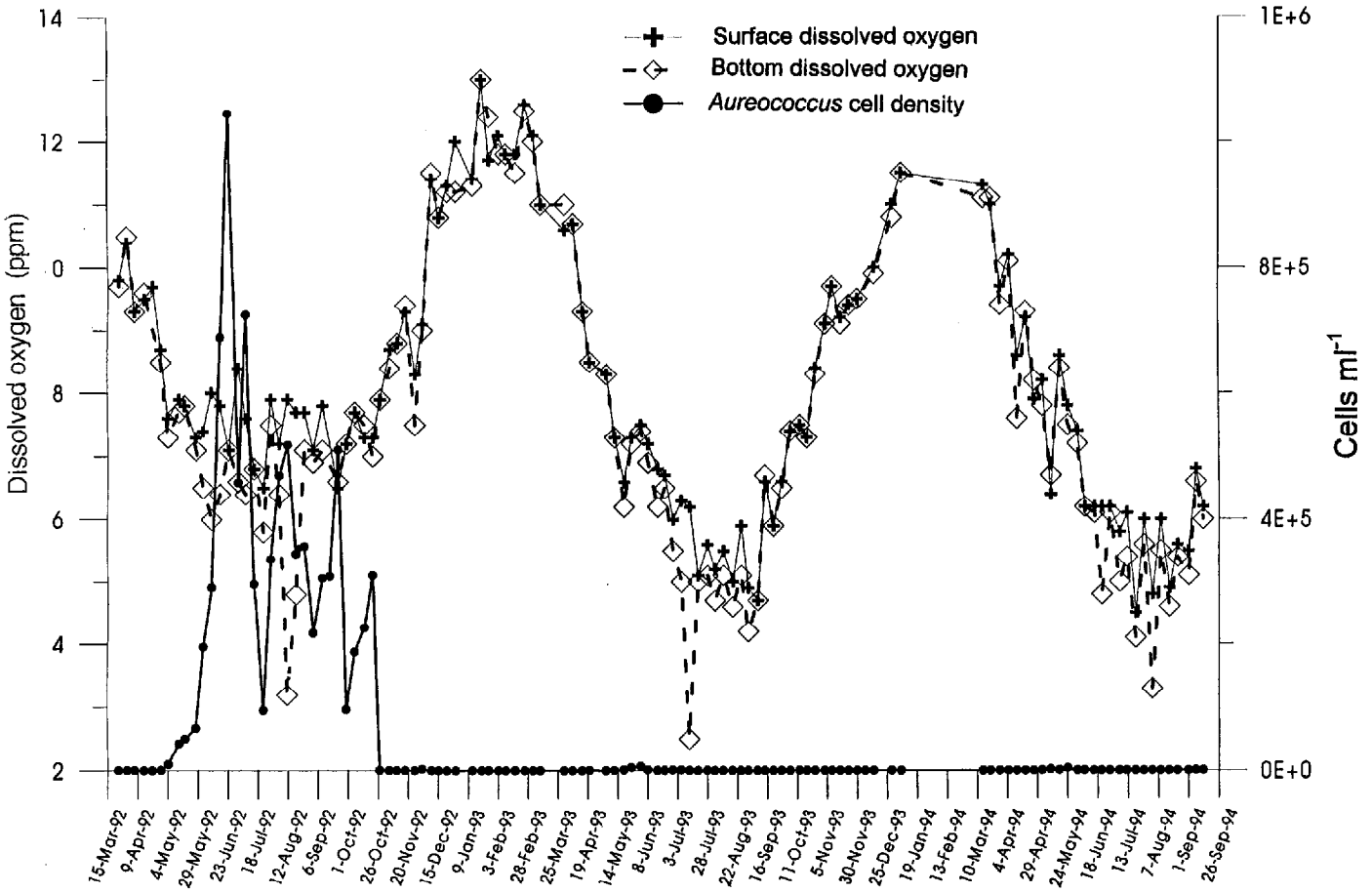

Fig. 6. Seasonal dissolved oxygen concentrations in surface waters and 1 m above the bottom in West Neck Bay in relation to cell densities of *Aureococcus anophagefferens*. Data are shown during a brown tide year (1992) and two years (1993, 1994) when brown tide did not develop at this site (data from R. Nuzzi, SCDHS; oxygen readings taken with YSI model 57 meter; station depth—3.7–4.3 m).

et al. 1987) and gonadal index—a measure of reproductive output (Bricelj and Kuenstner 1989)—in Peconic bays.

Unusually high mortalities (up to 64–82%) of adult bay scallops, measured from the incidence of articulated "clucker" shells, were determined immediately after the 1995 brown tide in Peconic Bay sites where *Aureococcus* densities reached $0.8–1.1 \times 10^6$ cells ml^{-1} (C. Smith pers. comm.). However, mortality data at these sites before the 1995 brown-tide outbreak, or during nonbrown-tide years, are lacking for comparison. Overall, economic losses for the New York State bay scallop fishery resulting from reduced landings attributed to brown tides (Fig. 8) were estimated at $2 million per year (Kahn and Rockel 1988). Stock enhancement practices, i.e. bottom plantings of hatchery-reared juveniles (Tettelbach and Wenczel 1993) have contributed to the partial recovery of scallop stocks in Peconic bays, as demonstrated using electrophoretic markers (Krause unpubl.). Although effects on other bivalves, such as oysters and quahogs remain poorly documented, poor growth of juvenile *M. mercenaria* during brown tides was reported by several commercial hatcheries (e.g. Bluepoints Inc., W. Sayville, New York, and Biosphere Inc., Tuckerton, New Jersey).

Adverse effects of brown tide on bivalve growth have not been linked to poor algal nutritional value, as measured by the complement of essential polyunsaturated fatty acids (PUFA) and digestibility of algal cells. Lipids of *Aureococcus* contain essential PUFA such as 20:5n-3 and 22:6n-3 at levels comparable to those of algal species of high nutritional value (Bricelj et al. 1989). Furthermore, the presence of an alternate food source in a mixed suspension with *A. anophagefferens* did not mitigate adverse effects on larval growth (Fig. 7A), as expected if nutritional deficiency were the primary cause of growth inhibition. Radiolabeling techniques have also shown that adult mussels and bay scallops absorb ingested *Aureococcus* cells with high efficiency (91–92%) (Bricelj and Kuenstner 1989) and that bay scallop larvae absorb brown-tide cells at levels comparable to good algal diets (Gallager et al. 1989). Thus, *Aureococcus* differs in its mode of action from other picoplanktonic algae such as *Nannochloris atomus* and *Stichococcus* sp., which formed dense blooms in Great South Bay in the 1950s. These algae result in brief gut transit times and are poorly absorbed by bivalves and are thereby unable to support bivalve growth (Bricelj et al. 1984; Bass et al. 1990).

Some studies indicate that there are marked differences among bivalve species in susceptibility to brown tide. *Aureococcus* inhibited gill ciliary activity in some species (*M.*

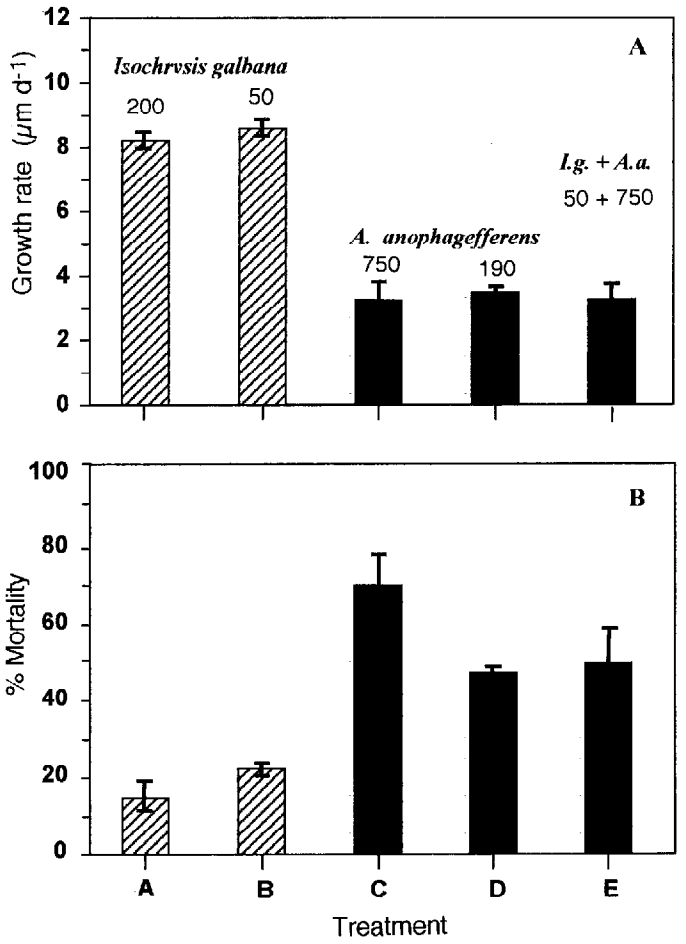

Fig. 7. Shell growth (A) and % mortality (B) of bay scallop (*Argopecten irradians*) larvae (3–10 d of development) fed unialgal and mixed cultured diets of *Aureococcus anophagefferens* (*A.a.*) and *Isochrysis galbana* (*I.g.*) (redrawn from Gallager et al. 1989). Solid bars indicate treatments with *Aureococcus*; cell densities, in cells liter^{-1} are shown above each bar.

mercenaria, M. edulis, Modiolus modiolus and the oysters *Ostrea edulis* and *Crassostrea virginica*) but had no effect on others (*Geukensia demissa* and *A. irradians*—a species known to be adversely affected by brown tides) (Gainey and Shumway 1991). Such discrepancies between in vitro studies using cultured *Aureococcus* and field observations on whole organisms need to be resolved. Growth of juvenile *M. mercenaria* and *M. edulis* held in suspended nets at several locations within the Peconics Bay system were compared during a moderate brown-tide outbreak in 1991 (Fig. 9). At densities of $1–3\times10^5$ *Aureococcus* cells ml^{-1}, mussels experienced much stronger growth reduction than quahogs, relative to a nonbloom site where *Aureococcus* remained at 5×10^3 cells ml^{-1}. Among-site growth differences could not be attributed to temperature differences, which were generally small (<0.5°C for all sites other than Flanders Bay). This and other studies also indicate that bivalves may experience sublethal, adverse effects at moderate *Aureococcus* cell densities (10^5 cells ml^{-1}). Levels exceeding $\sim2.5\times10^5$ cells ml^{-1} were required to inhibit clearance rates of mussels on *I. galbana*, a good algal food source (Fig. 5; Tracey 1988). In the laboratory, *Aureococcus* densities $>1.9\times10^5$ cells ml^{-1} resulted in a significant decrease in growth and increase in mortality of *A. irradians* larvae (Fig. 7; Gallager et al. 1989). Bricelj et al. (1987) found that growth of surviving adult *A. irradians* did not resume until field concentrations of *Aureococcus* dropped below $\sim2.3\times10^5$ cells ml^{-1}. Further work is necessary to define species- and stage-specific effects as a function of *Aureococcus* concentration, duration of exposure, and cell toxicity.

Planktonic—In a field study in Long Island bays, Caron et al. (1989) found that densities of bacteria, ciliates, and heterotrophic nanoplankton were not correlated with the density of *Aureococcus* and not negatively affected by moderate concentrations (up to $\sim5\times10^5$ cells ml^{-1}). Moreover, across a broad range of cell concentrations (from nonbloom to bloom levels), brown tide also had no obvious effect on the composition of the heterotrophic microplankton. At higher *A. anophagefferens* densities, however, detrimental impacts on protozoa have been found. Brown tide (at $\sim1.5\times10^6$ cells ml^{-1}) in West Neck Bay in 1991 was associated with negative growth rates of both aloricate ciliates and tintinnids (-1.4 d^{-1}; Lonsdale et al. 1996). As brown tide declined to 5×10^5 cells ml^{-1}, population growth rates of aloricate ciliates recovered (0.3 d^{-1}) and at 3.0×10^4 cells ml^{-1} were equivalent to those under summer, nonbloom conditions (1.2 d^{-1}). The composition of ciliates, however, changed notably after the decline of brown tide: the population was composed almost exclusively of small ($\sim30–40$ μm), aloricate ciliates, but no tintinnids—a common component of the ciliate assemblage under nonbloom conditions.

Negative impacts on ciliates were again recorded in 1995 in West Neck Bay. From the onset of brown tide to its peak (at 1.1×10^6 cells ml^{-1}), the microzooplankton population declined from >10,000 to <900 ind. liter^{-1} (Mehran 1996). Rapid recovery of aloricate ciliates (>4,000 ind. liter^{-1}), but not of tintinnids, was again observed during brown tide dissipation. The reduction in ciliate density may be related to a reduced nutritional capacity of *A. anophagefferens* to support ciliate growth. Although *Strombidium* sp. ingested cultured brown tide cells, its population growth was significantly lower when fed a diet of *Aureococcus* (isolate CCMP1708 at 2×10^6 cells ml^{-1}) compared to a diet of *I. galbana*, but with mixed phytoplankton (e.g. equal carbon rations of the two species) the negative impact of *Aureococcus* on population growth was not apparent (Mehran 1996).

Taken together, these studies (Caron et al. 1989; Lonsdale et al. 1996; Mehran 1996) suggest a "threshold" phenomenon in which predator-prey interactions, especially protozoan grazing and production, seem little effected over a wide range of *Aureococcus* cell concentrations and are disrupted only during peak bloom conditions or when alternate food is lacking. The impacts of brown tide on protozoa are therefore expected to strongly reflect the relative availability of other food sources. A prolonged period of unialgal high concentrations ($>1.0\times10^6$ cells ml^{-1}) of *Aureococcus* is likely to have widespread, detrimental effects on production rates of higher trophic levels via its impacts on protozooplankton.

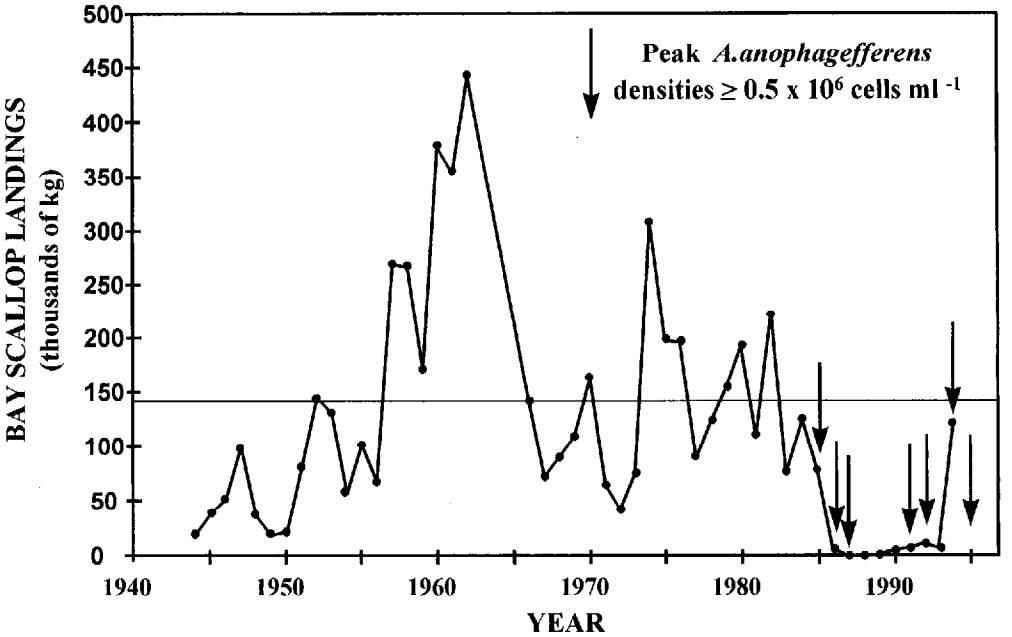

Fig. 8. Effect of *Aureococcus anophagefferens* outbreaks on landings of northern bay scallops, *Argopecten irradians irradians,* in New York State (horizontal line indicates the mean historical annual production level prior to the brown tide; landing records from T. Drumm, New York State Dep. Environ. Conserv.). (Note that *Aureococcus* densities reached 0.56×10^6 cells ml^{-1} in West Neck Bay in 1990.)

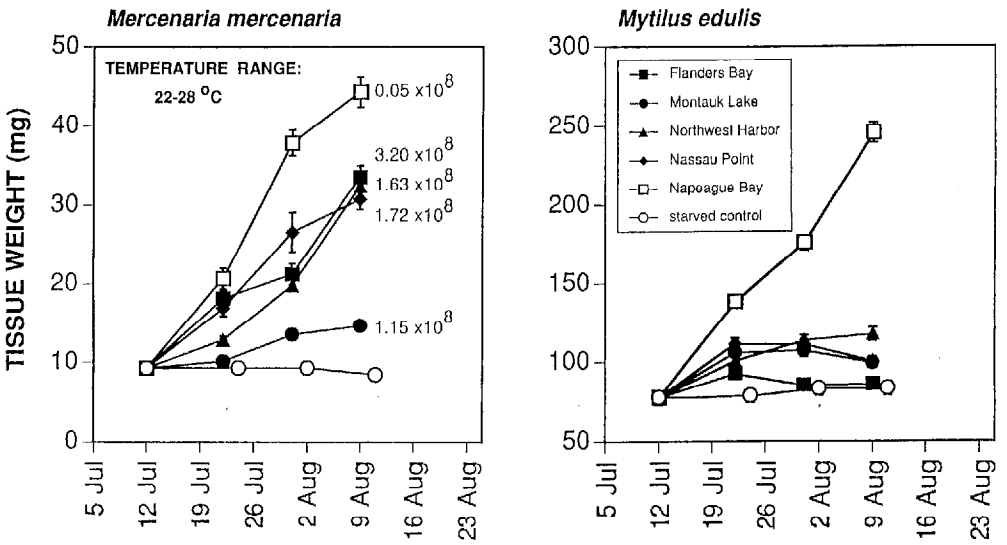

Fig. 9. Mean dry weight of soft tissues (± SE) of juvenile quahogs (*M. mercenaria*) and blue mussels (*M. edulis*) suspended in pearl nets at each of five sites (*see Fig. 1*) in Peconic-Gardiners bays during a 1991 summer bloom of *Aureococcus anophagefferens,* and of starved controls held in 0.22-μm-filtered seawater in the laboratory at ambient temperature. Numbers indicate mean cell densities of *Aureococcus* determined by immunofluorescence (SCDHS) during the study period (cells $liter^{-1}$) (Bricelj and Borrero unpubl. data). Cumulative mortalities of quahogs were negligible (<1%) at all locations, including controls; those of mussels were <5% at all sites except Flanders Bay, where they reached 19%

Such trophic-level impacts have been found during the prolonged Texas brown tide (Buskey and Stockwell 1993).

The importance of ciliates in the diets of larger zooplankton (*see* Stoecker and Cappuzo 1990), especially for adult copepods in summer coastal waters (e.g. Gifford and Dagg 1991; Lonsdale et al. 1996), and the concentration-related effects of brown tide on ciliate population growth may explain why abundances of copepod populations in Great South Bay during brown tides (1985 and 1986) were like those found during nonbloom years (Duguay et al. 1989). No data are available for 1985 on phytoplankton species composition in Great South Bay. In 1986, when extensive monitoring of the brown tide was conducted in Great South Bay, the average concentration was 1.4×10^5 cells ml^{-1} and reached a peak of only $6–7\times10^5$ cells ml^{-1} (Nuzzi and Waters 1989). Moreover, other phytoplankton such as *Nannochloris* sp.—a likely food resource for many protozoa—outnumbered *A. anophagefferens*. Thus, microbial food-web processes likely remained intact and allowed for normal zooplankton productivity levels.

Durbin and Durbin (1989), however, found that during the Narragansett Bay brown tide, copepod production was reduced at 7.6×10^5 *Aureococcus* cells ml^{-1}. They reported that *A. tonsa* weight, condition, and egg production rate during 1985 were very low, although like those sometimes found in nonbloom years in summer when food-limiting conditions normally exist. In contrast to findings for Great South Bay, there was an inverse correlation between *Aureococcus* cell concentration and the abundance of *A. tonsa* in Narragansett Bay, suggesting negative impacts on secondary production (Smayda and Villareal 1989). Preliminary data showed that copepod egg production rates in West Neck Bay were also negatively impacted after the brown tide had reached a concentration of 1.5×10^6 cells ml^{-1} (Lonsdale et al. 1996). Adverse effects on *A. tonsa* reproduction were similarly described during the Texas brown tide (Buskey and Stockwell 1993).

In the laboratory, no detrimental effects of cultured *Aureococcus* at a concentration of 5×10^5 cells ml^{-1} on naupliar and copepodite survival were observed when alternate food was available (Fig. 10). These results contrast with those described earlier (Gallager et al. 1989) for scallop larvae in which the addition of alternate phytoplankton to brown tide failed to mitigate negative effects (Fig. 7). The laboratory study also showed that a monospecific diet of *A. anophagefferens* is inadequate for copepod development, as naupliar and copepodite survival were similar to that in only filtered seawater. Thus the impact of *Aureococcus* on copepods is expected to be largely dependent on the availability of alternate food during brown tides.

In contrast to the "mixed" findings on the impacts of brown tide on copepod populations, other zooplankton taxa were clearly adversely affected by its presence. In Narragansett Bay, the annual cycle of high summer densities of cladocerans, including *Evadne nordmanni* and *Podon* sp., were notably absent during the 1985 brown tide. In 1986, when normal conditions prevailed, abundances returned to levels found in previous nonbloom years (Smayda and Fofonoff 1989). Given that some cladocerans preferentially ingest small (2–5 μm) algae comparable in size to brown tide,

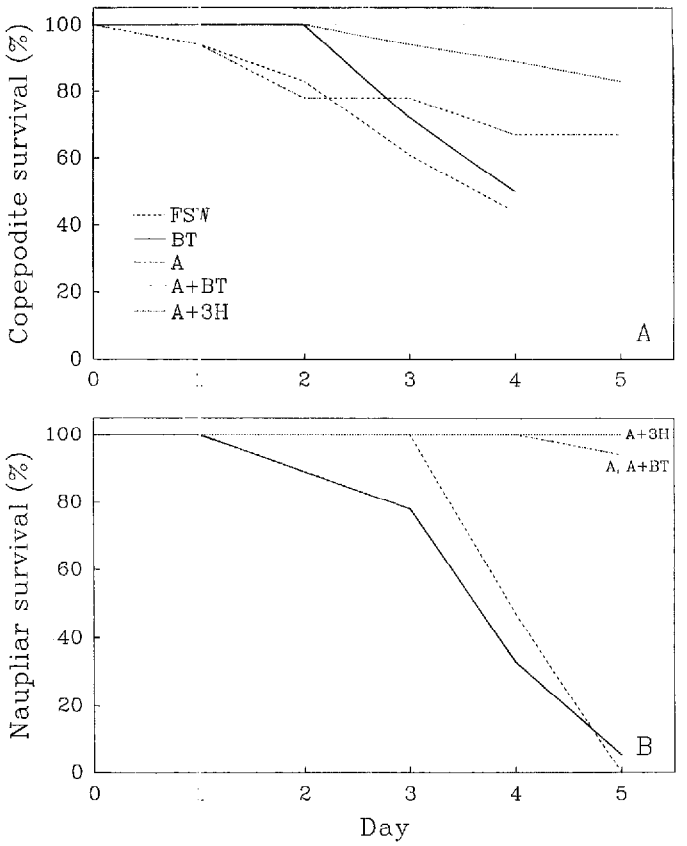

Fig. 10. Percent survival of developmental stages of laboratory-reared copepods that commonly occur in Long Island bays. *A. Acartia hudsonica* (N6–C3); B. *Coullana canadensis* (N1–C1). Copepods were reared at 16° and 20°C, respectively, and five food treatments (BT—cultured brown tide cells at 5×10^5 cells ml^{-1}; FSW—filtered seawater; A—ambient seawater; A+BT—ambient plus *Aureococcus* cells; A+3H—ambient plus cultured *Thalassiosira pseudonana* cells) (from Lonsdale et al. 1996). Additions of cultured microalgae were equivalent to a carbon concentration of 1,100 μg C $liter^{-1}$ according to the equations of Strathmann (1967).

and not larger or chain-forming diatoms (Turner et al. 1988), cladoceran grazing may have been seriously impacted by brown tide. In Narragansett Bay, abundances of meroplanktonic larvae, including polychaetes and bivalves, were also negatively correlated with brown tide concentration and were lower than in nonbloom years (Smayda and Fofonoff 1989). Duguay et al. (1989) also found decreased abundances of bivalve larvae (attributed primarily to *M. mercenaria*) during the 1985 bloom event in Great South Bay, but not during 1986—a year of less severe brown tide—and speculated that there might be an association between brown tide levels and quahog spawning success or larval survival.

Concentration-related effects of brown tide on microzooplankton may help explain why larval fish growth was unaffected at lower concentrations of brown tide (1987 and 1988; Castro and Cowen 1989) given that microzooplankton serve as food for larval fish. In addition to potential indirect food-web effects, increased turbidity caused by brown tide may affect larval fish by reducing their ability to capture prey and avoid predators. In mid-Atlantic estuaries, includ-

ing Long Island bays, bay anchovy (*Anchoa mitchilli*) larvae, which provide an important food source for bluefish and striped bass, dominate the ichthyoplankton in summer, when brown tide occurs (Duguay et al. 1989; Shima and Cowen 1989). Yet larval bay anchovy abundance and growth were not affected during two moderate brown tide years (1987 and 1988) in Great South Bay relative to areas unaffected by brown tide (Shima and Cowen 1989; Castro and Cowen 1989). No comparative data are available, however, for prebrown tide years or more intense blooms. This lack of documented effects on larval fish contrasts markedly with the severe declines in egg hatching rate and survival of fish larvae associated with the Texas brown tide (T. Whitledge unpubl. data). It is noteworthy, however, that mean egg abundance of *A. mitchilli* during the 1985 Narragansett Bay brown tide was only 10% of mean levels in 1973 and 1981–1986 (Smayda and Villareal 1989).

Conclusions

Several environmental conditions are conducive for development of *Aureococcus* blooms in shallow estuaries: reduced flushing rates, elevated water salinities (≥26–28‰), and temperatures. Present work indicates that brown tides do not develop in response to inorganic macronutrient (N and P) enrichment. Iron, however, has been implicated as an important stimulatory factor for *Aureococcus* growth, but its bioavailability and role in controlling brown tide dynamics remain unknown. The ability of *Aureococcus* to outcompete other phytoplankton species was not related to allelopathic effects on other microalgae but may be associated with its capacity to take up organic nurients and photoadapt to low light levels. Despite these findings, the combination of factors which triggers the onset of brown tide (e.g. availability of stimulatory micronutrients or organic macronutrients via freshwater input or sediment resuspension, physical forcing resulting from rapidly increasing temperatures, or reduced grazing pressure) is not well understood. Current hypotheses are unable to account for the high degree of spatial and interannual variability of brown tide in Long Island estuaries. An important caveat in attempts to extrapolate laboratory results reported here to the field is that these were largely obtained with nonaxenic cultures of a single isolate of *A. anophagefferens* (isolated by E. Cosper in 1986). Other isolates (e.g. CCMP1708) have become available only recently.

Reduced grazing pressure on *Aureococcus,* relative to other picoplanktonic algae, was demonstrated in a number of suspension-feeders: meroplankton (bivalve larvae), macrofauna (adult bivalves), and microzooplankton (primarily protozoans), typically the dominant planktonic primary consumer in these bays in summer. Therefore, reduced grazing by planktonic and benthic communities clearly contributes to the maintenance of brown tide in shallow bays. Little is known, however, about grazing pressure at low *Aureococcus* densities and its potential to curtail bloom initiation or the degree to which grazing (albeit depressed) contributes to bloom decline. In Long Island bays poor grazing control of *Aureococcus* at low to moderate densities may be exacerbated, via negative feedback effects, by the recent declines of bivalve populations caused by brown tides (e.g. of *A. irradians* and *C. virginica*), overharvesting (*M. mercenaria*) and (or) other causes.

In affected waters, brown tide causes severe light attenuation, and thereby reduction of eelgrass cover, but has not led to hypoxic-anoxic conditions. Although brown tide does not pose a direct human health hazard, and effects on secondary consumers (e.g. finfish and decapods) have so far not been reported, its impact on benthic and planktonic herbivores (especially bivalve molluscs and microzooplankton) is well documented. Mass mortalities of bivalves and declines in protozoan populations have occurred during blooms. Adverse effects of brown tide on feeding and growth of suspension-feeders were shown to vary both among and within taxa and do not seems to be explained by a common underlying mechanism. Whereas *A. anophagefferens* supports growth of some protozoans, it is nutritionally inadequate for others (e.g. *Strombidium* sp.) and for copepod developmental stages. *Aureococcus* lies below the optimum size range for particle capture of some of the larger, metazoan suspension-feeders present during summer in estuaries affected by brown tide, namely larval anchovy, adult copepods such as *A. tonsa,* and postsettlement bivalves. Reduced retention efficiency of picoplankton by bivalves could, however, be offset by their ability to process large volumes of water. Yet *Aureococcus,* at moderate to high densities, seems to be toxic to at least some bivalve larvae and adults, in which it elicits nonselective, postcapture rejection of microalgae, and inhibition of ciliary feeding currents, respectively. The putative toxin involved remains uncharacterized.

Both the absolute and relative abundance of *Aureococcus* can be critical in determining its impact on grazers. Adverse effects on bivalve larvae and adults seem to be induced above a threshold concentration of $\sim 2\times10^5$ *Aureococcus* cells ml^{-1}. Existing data suggest that *Aureococcus* inhibits grazing on other co-occurring phytoplankton by bivalves (scallop larvae and adult mussels) but not by microzooplankton (copepod nauplii and protozooplankton), some of which can selectively avoid *Aureococcus* cells. Thus the impact of brown tide on microzooplankton seems to be more dependent on the relative rather than absolute density of *Aureococcus* and is mitigated, in contrast to mussels, by the presence of alternate food in a mixed assemblage.

In general, benthic bivalve communities and bivalve larvae (meroplankton) seem to be more "sensitive" compared to holoplanktonic communities to brown tides, at least as measured by cell concentration-dependent impacts, the ability to tolerate *Aureococcus* in mixed phytoplankton assemblages, and time required for restoration of populations to prebloom conditions. Sensitivity differences among benthic and planktonic communities are likely due, in part, to variation in food-web structure. Benthic macrofauna present in estuaries affected by brown tide are primarily herbivores and thus part of the traditional food, while during summer, when brown tides prevail, some zooplankters are more dependent on "microbial loop" processes (e.g. feeding on protozoa or bacteria). Marked differences in susceptibility to brown tide were also found within taxa: among bivalve species *M. edulis* seems to be more vulnerable than *M. mercenaria,* and among protozoans tintinnids are more severely affected than

aloricate ciliates. Thus brown tide can potentially cause significant changes in the species composition and trophic structure of suspension-feeding communities in shallow estuaries.

References

ANDERSON, D. M., B. A. KEAFER, D. M. KULIS, R. M. WATER, AND R. NUZZI. 1993. An immunofluorescent survey of the brown tide chrysophyte *Aureococcus anophagefferens* along the northeast coast of the United States. J. Plankton Res. **15:** 563–580.

———, D. M. KULIS, C. M. CETTA, AND E. M. COSPER. 1989. Immunofluorescent detection of the brown tide organism, *Aureococcus anophagefferens,* p. 213–228. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

BASS, A. E., R. E. MALOUF, AND S. E. SHUMWAY. 1990. Growth of northern quahogs (*Mercenaria mercenaria* (Linnaeus, 1758) fed on picoplankton. J. Shellfish Res. **9:** 299–307.

BRICELJ, V. M., A. E. BASS, AND G. R. LOPEZ. 1984. Absorption and gut passage time of microalgae in a suspension-feeder: An evaluation of the $^{51}Cr:^{14}C$ twin tracer technique. Mar. Ecol. Prog. Ser. **17:** 57–63.

———, J. EPP, AND R. E. MALOUF. 1987. Intraspecific variation in reproductive and somatic growth cycles of bay scallops *Argopecten irradians.* Mar. Ecol. Prog. Ser. **36:** 123–137.

BRICELJ, V. M., N. S. FISHER, J. B. GUCKERT, AND F.-L. E. CHU. 1989. Lipid composition and nutritional value of the brown tide alga *Aureococcus anophagefferens,* p. 85–100. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

———, AND S. H. KUENSTNER. 1989. Effects of the "brown tide" on the feeding physiology and growth of bay scallops and mussels, p. 491–509. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

BUSKEY, E. J., AND D. A. STOCKWELL. 1993. Effects of a persistent "brown tide" on zooplankton populations in the Laguna Madre of South Texas, p. 659–665. *In* Toxic phytoplankton blooms in the sea. Proc. 5th Int. Conf. on Toxic Marine Phytoplankton. Elsevier.

CARON, D. A., E. LIM, H. KUNZE, E. M. COSPER, AND D. M. ANDERSON. 1989. Trophic interactions between nano- and microzooplankton and the "brown tide," p. 265–294. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

CASTRO, L. R., AND R. K. COWEN. 1989. Growth rates of bay anchovy (*Anchoa mitchilli*) in Great South Bay under recurrent brown tide conditions, summers 1987 and 1988, p. 663–674. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

COSPER, E. M., E. J. CARPENTER, AND M. COTTRELL. 1989*a.* Primary productivity and growth dynamics of the "brown tide" in Long Island embayments, p. 139–158. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

———, AND OTHERS. 1987. Recurrent and persistent brown tide blooms perturb coastal marine ecosystem. Estuaries **10:** 284–290.

———, AND OTHERS. 1989*b.* An examination of the environmental factors important to initiating and sustaining "brown tide" blooms, p. 317–340. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

———, R. T. GARRY, A. J. MILLIGAN, AND M. H. DOALL. 1993. Iron, selenium and citric acid are critical to the growth of the "brown tide" microalga, *Aureococcus anophagefferens,* p. 667–673. *In* Toxic phytoplankton blooms in the sea. Proc. 5th Int. Conf. on Toxic Marine Phytoplankton. Elsevier.

———, C. LEE, AND E. J. CARPENTER. 1990. Novel "brown tide" blooms in Long Island embayments: A search for the causes, p. 17–28. *In* Toxic marine phytoplankton: Proc. 4th Int. Conf. Elsevier.

DENNISON, W. C. 1987. Effects of light on seagrass photosynthesis, growth and depth distribution. Aquat. Bot. **27:** 15–26.

———, G. J. MARSHALL, AND C. WIGAND. 1989. Effect of "brown tide" shading on eelgrass (*Zostera marina* L.) distributions, p. 675–692. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

———, AND OTHERS. 1993. Assessing water quality with submersed aquatic vegetation. BioScience **43:** 86–94.

DEYOE, H. R., A. M. CHAN, AND C. A. SUTTLE. 1995. Phylogeny of *Aureococcus anophagefferens* and a morphologically similar bloom-forming alga from Texas as determined by 18S ribosomal RNA sequence analysis. J. Phycol. **31:** 413–418.

———, AND C. A. SUTTLE. 1994. The inability of the Texas "brown tide" alga to use nitrate and the role of nitrogen in the initiation of a persistent bloom of this organism. J. Phycol. **30:** 800–806.

DRAPER, C., L. GAINEY, S. SHUMWAY, AND L. SHAPIRO. 1990. Effects of *Aureococcus anophagefferens* ("brown tide") on the lateral cilia of 5 species of bivalve molluscs, p. 128–131. *In* Toxic marine phytoplankton: Proc. 4th Int. Conf. Elsevier.

DUGUAY, L. E., D. M. MONTELEONE, AND C. QUAGLIETTA. 1989. Abundance and distribution of zooplankton and ichthyoplankton in Great South Bay, New York, p. 599–623. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

DURBIN, A. G., AND E. G. DURBIN. 1989. Effect of the "brown tide" on feeding, size and egg laying rate of adult female *Acartia tonsa,* p. 625–645. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

DZURICA, S., C. LEE, E. M. COSPER, AND E. J. CARPENTER. 1989. Role of environmental variables, specifically organic compounds and micronutrients, in the growth of the chrysophyte *Aureococcus anophagefferens,* p. 229–252. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

GAINEY, L. F., JR., AND S. E. SHUMWAY. 1991. The physiological effect of *Aureococcus anophagefferens* ("brown tide") on the lateral cilia of bivalve mollusks. Biol. Bull. **181:** 298–306.

GALLAGER, S. M. 1988. Visual observations of particle manipulation during feeding in larvae of a bivalve mollusc. Bull. Mar. Sci. **43:** 344–365.

———, D. K. STOECKER, AND V. M. BRICELJ. 1989. Effects of the brown tide alga on growth, feeding physiology and locomotory behavior of scallop larvae (*Argopecten irradians*), p. 511–541. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

———, J. B. WATERBURY, AND D. K. STOECKER. 1994. Efficient grazing and utilization of the marine cyanobacterium *Synechococcus* sp. by larvae of the bivalve *Mercenaria mercenaria.* Mar. Biol. **11:** 251–259.

GIFFORD, D. J., AND M. J. DAGG. 1991. The microzooplankton–mesozooplankton link: Consumption of planktonic protozoa by the calanoid copepods *Acartia tonsa* Dana and *Neocalanus plumchrus* Murukawa. Mar. Microbiol. Food Webs **5:** 161–177.

GOBLER, C. J. 1995. The role of iron in the occurrence of *Aureococcus anophagefferens* blooms. M.S. thesis, SUNY Stony Brook. 127 p.

———, AND E. M. COSPER. 1996. Stimulation of "brown tide" blooms by iron, p. 321–324. *In* Harmful and toxic algal blooms. Proc. 7th Int. Conf. on Toxic Phytoplankton. Intergov. Oceanogr. Comm. UNESCO.

GUILLARD, R. R. L., AND J. H. RYTHER. 1962. Studies of marine planktonic diatoms. 1. *Cyclotella nana* Hustedt and *Detonula confervacea* (Cleve) Gran. Can. J. Microbiol. **8:** 229–239.

KAHN, J., AND M. ROCKEL. 1988. Measuring the economic effects of brown tides. J. Shellfish Res. **7:** 677–682.

KELLER, A. A., AND R. L. RICE. 1989. Effects of nutrient enrichment on natural populations of the brown tide phytoplankton *Aureococcus anophagefferens* (Chrysophyceae). J. Phycol. **25:** 636–646.

KELLER, M. D., W. K. BELLOWS, AND R. R. L. GUILLARD. 1989. Dimethylsulfide production and marine phytoplankton: An additional impact of unusual blooms, p. 101–115. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

LONSDALE, D. J., AND OTHERS. 1996. Food web interactions in the plankton of Long Island bays, with preliminary observations on brown tide effects. Mar. Ecol. Prog. Ser. **134:** 247–263.

MEHRAN, R. 1996. Effects of *Aureococcus anophagefferens* on microzooplankton grazing and growth rates in the Peconic Bays system, Long Island, NY. M.S. thesis, SUNY Stony Brook.

MILLIGAN, A. J. 1992. An investigation of factors contributing to blooms of the "brown tide" *Aureococcus anophagefferens* (Chrysophycea) under nutrient saturated, light limited conditions. M.S. thesis, SUNY Stony Brook. 84 p.

MILLIGAN, K. L. D., AND E. M. COSPER. 1994. Isolation of virus capable of lysing the brown tide microalga, *Aureococcus anophagefferens*. Science **266:** 805–807.

MØHLENBERG, F., AND H. U. RIISGARD. 1978. Efficiency of particle retention in 13 species of suspension feeding bivalves. Ophelia **17:** 239–246.

MONTAGNA, P. A., D. A. STOCKWELL, AND R. D. KALKE. 1993. Dwarf surfclam *Mulinia lateralis* (Say, 1822) populations and feeding during the Texas brown tide event. J. Shellfish Res. **12:** 433–442.

NIXON, S. W., S. L. GRANGER, D. I. TAYLOR, P. W. JOHNSON, AND B. A. BUCKLEY. 1994. Subtidal volume fluxes, nutrient inputs and the brown tide—an alternate hypothesis. Estuarine Coastal Shelf Sci. **39:** 303–312.

NUZZI, R., AND R. M. WATERS. 1989. The spatial and temporal distribution of "brown tide" in eastern Long Island Sound, p. 117–138. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

OLSEN, P. S. 1989. Development and distribution of a brown-water algal bloom in Barnegat Bay, New Jersey with perspective on resources and other red tides in the region, p. 189–212. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

PALENIK, B. P., AND F. M. M. MOREL. 1990. Amino acid utilization by marine phytoplankton: A novel mechanism. Limnol. Oceanogr. **35:** 260–269.

PANTOJA, S. 1992. Measurement of cell-surface enzymatic oxidation of amino acids and amines in seawater using synthetic fluorescent compounds. M.S. thesis, SUNY Stony Brook. 128 p.

———, AND C. LEE. 1994. Cell-surface oxidation of amino acids in seawater. Limnol. Oceanogr. **39:** 1718–1726.

PETERSON, C. H., AND H. C. SUMMERSON. 1992. Basin scale-coherence of population dynamics of an exploited marine invertebrate, the bay scallop: Implications of recruitment limitation. Mar. Ecol. Prog. Ser. **90:** 257–272.

POHLE, D. G., V. M. BRICELJ, AND Z. GARCIA-ESQUIVEL. 1991. The eelgrass canopy: An above-bottom refuge from benthic predators for juvenile bay scallops *Argopecten irradians*. Mar. Ecol. Prog. Ser. **74:** 47–59.

SCHNEIDER, T. R. 1994. Break in "brown tide" studies. Coastlines **24:** 11–12.

SHIMA, M., AND R. K. COWEN. 1989. Potential change in the distribution of larval fish within Great South Bay, New York in response to recurrent phytoplankton blooms, p. 647–662. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

SHORT, F. T., D. M. BURDICK, AND J. E. KALDY III. 1995. Mesocosm experiments quantify the effects of eutrophication on eelgrass, *Zostera marina*. Limnol. Oceanogr. **40:** 740–749.

SIEBURTH, J. MCN., AND P. W. JOHNSON. 1989. Picoplankton ultrastructure: A decade of preparation for the brown tide alga, *Aureococcus anophagefferens*, p. 1–22. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

———, ———, AND P. E. HARGRAVES. 1988. Ultrastructure and ecology of *Aureococcus anophagefferens* gen. et sp. nov. (Chrysophyceae); the dominant picoplankter during a bloom in Narragansett Bay, Rhode Island, Summer 1985. J. Phycol. **24:** 416–425.

SIEGEL, D. I., A. S. REEVE, P. H. GLASER, AND E. A. ROMANOWICZ. 1995. Climate-driven flushing of pore water in peatlands. Nature **374:** 531–533.

SMAYDA, T. J., AND P. FOFONOFF. 1989. An extraordinary, noxious brown-tide in Narragansett Bay. 2. Inimical effects, p. 133–136. *In* Red tides: Biology, environmental science, and toxicology. Proc. 1st Int. Symp. on Red Tides. Elsevier.

———, AND T. A. VILLAREAL. 1989. The 1985 "brown-tide" and the open phytoplankton niche in Narragansett Bay during summer, p. 159–188. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

STOECKER, D. K., AND J. M. CAPUZZO. 1990. Predation on protozoa: Its importance to zooplankton. J. Plankton Res. **12:** 891–908.

STRATHMANN, R. R. 1967. Estimating the organic carbon content of phytoplankton from cell volume or plasma volume. Limnol. Oceanogr. **12:** 411–418.

TETTELBACH, S. T., AND P. WENCZEL. 1993. Reseeding efforts and the status of bay scallop *Argopecten irradians* (Lamarck, 1819) populations in New York following the occurrence of "brown tide" algal blooms. J. Shellfish Res. **12:** 423–431.

TRACEY, G. A. 1988. Feeding reduction, reproductive failure, and mortality in *Mytilus edulis* during the 1985 "brown tide" in Narragansett Bay, Rhode Island. Mar. Ecol. Prog. Ser. **50:** 73–81.

———, P. W. JOHNSON, R. W. STEELE, P. E. HARGRAVES, AND J. MCN. SIEBURTH. 1988. A shift in photosynthetic picoplankton composition and its effect on bivalve mollusc nutrition: The 1985 "brown tide" in Narragansett Bay, Rhode Island. J. Shellfish Res. **7:** 671–675.

———, R. STEELE, AND L. WRIGHT. 1990. Variable toxicity of the brown tide organism, *Aureococcus anophagefferens*, in relation to environmental conditions for growth, p. 233–237. *In* Toxic marine phytoplankton: Proc. 4th Int. Conf. Elsevier.

TURNER, J. T., P. A. TESTER, AND R. L. FERGUSON. 1988. The marine cladoceran *Penilia avirostris* and the "microbial loop" of pelagic food webs. Limnol. Oceanogr. **33:** 245–255.

VIEIRA, M. E. C. 1989. The case for meterologically driven fluctuations in residence times of Long Island waters subject to algal blooms, p. 295–306. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

———, AND R. CHANT. 1993. On the contribution of subtidal volume fluxes to algal blooms in Long Island estuaries. Estuarine Coastal Shelf Sci. **36:** 15–29.

WARD, J. E., AND N. M. TARGETT. 1989. Are metabolites from the brown tide alga, *Aureococcus anophagefferens,* deleterious to mussel feeding behavior?, p. 543–556. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.

YENTSCH, C. S., D. A. PHINNEY, AND L. P. SHAPIRO. 1989. Absorption and fluorescent characteristics of the brown tide chrysophyte. Its role on light reduction in coastal marine environments, p. 77–84. *In* E. M. Cosper et al. [eds.], Novel phytoplankton blooms: Causes and impacts of recurrent brown tides and other unusual blooms. Springer.